ActInf Textbook Group ~ Cohort 2 ~ Meeting 17 (Chapter 7, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 17 (Chapter 7, part 2) | 54:35

Hello, it's February 22nd, 2023, 222, 23, and we're in the second discussion of Chapter 7 in Cohort 2.
Before we go to the questions of Chapter 7, and then at the end, a look ahead to Chapter 8,
we are going to have Thomas Parr join for a live stream in five days.
And so anybody, if they would like to raise an area of questions, could be conceptual,
or it could just be a general textbook level or personal question.
So of course, feel free to add anything, but just while we're here, any thought or like kind of general
piece that people are thinking that Thomas could reduce our uncertainty about?
All right.
Just, of course, add it when you see fit.
Okay.
There were several questions that we've already looked at from 7 and several more.
We can also look a little bit more.
Just again, we looked at the first sections of the chapter itself.
So, does anyone have a Chapter 7 question that they want to go to or add?
Or do they prefer to walk through the sections in the textbook?
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Okay.
Let's look at equation 7.4.
See if we have anything annotated for it.
Okay.
Okay.
We don't have the natural language yet.
None of the natural language yet.
Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

Okay.

So, let's go back up with the rat in the maze example.

Again, go through it.

Connect it to the equations and then return to several of the questions down here in 7.

So in the earliest part of chapter 7 was a passive inference with the example of listening to music.

Now the upstairs action selection part is going to come into play.

The imperative for action selection is going to be G. G, like we saw in chapter 5,

takes the action prior, the prior on different policies, which can be understood as like habit,

and then sharpens it according to, just qualitatively, provide epistemic and pragmatic value,

and quantitatively as described in equation 2.6 and 7.4.

Does anyone want to give a reading or a thought on 7.4?

Just what is it saying in and of itself?

And or how does or might this relate to, in their words, the resolution of the exploration-exploitation dilemma?

Yeah, Ali, go for it.

Well, yeah, as we saw before in chapter 2, I guess, but I think it was in more comprehensive terms.

But here we only have a part of that equation, which basically deals with the trade-off between negative,

I mean, epistemic value and the pragmatic value, and how they relate to each other.

Because as we saw in chapter 2, the trade-off or the dilemma between exploration and exploitation

is basically one of the fundamental requirements for any sentient behavior,

because as long as the agent tries to, wants to explore the environment in order to informally discover

new, let's say, opportunities or new resources, be it, I don't know, food resource or any other resource

necessary for its survival in the environment,

it needs to somehow have a trade-off between this kind of having the pragmatic value on one hand,

namely to survive in the environment, and on the other hand, to discover new resources for its survival,

which basically relates to the pragmatic value.

But either of those two requirements, if taken to its extreme far, would probably result in the demise of the agent,

because, so take, for instance, an agent residing in an environment, but for a specific,

a certain period of time, it can survive in that environment using the pre-known resources there.

But when the resources is about to,

is about to diminish,

then that agent needs to explore a little bit further in order to discover some new resources and continue its persistent through time or its survival.

Yeah. Great.

Yes, Michael.

So, I'm a little bit wondering, in this epistemic considerations,

only things can be considered that the agent actually already knows about, right?

And then I'm wondering why, at several places, I think I said this in chapter 2 and here in this chapter 2,

also 7 too, also that one would like to find the most unlikely path,

or one would like to increase the information value means one,

that's what I understood.

So, the path that should be searched for would be the path that maximizes information,

which means it's the most uncertain path.

And, I mean,

I'm incorrect.

There's aspects that I think are correct, but I just want to draw that out.

So, actually, what is being selected for policy are the most likely paths.

What is an agent like me most likely to do?

We're sampling our actions from the action posterior.

So, rather than scaling policies by their reward and ranking them by their reward

in reward learning and choosing sampling from the reward posterior, for example,

we're going to be sampling from an action posterior.

So, in the sense that our actions are selected based upon their posterior likelihood,

we're actually pursuing the most likely path of action, indeed the path of least action.

However, we are seeking in the epistemic component especially maximally informative sequences of future observations.

So, in the special case where pragmatic value doesn't matter, like we have no preference over outcomes,

we don't care if our bank account goes up or down.

Then, we implement a pure Infomax novelty search.

In a situation that's totally known, there's no epistemic value to be obtained,

we have a special case of pure utility pragmatic value.

However, in cases where both utility pragmatic and epistemic are on the table,

G is proposed to resolve this dilemma by adjusting the action prior into an action posterior

that respects the importance of both of those imperatives.

That's the unified imperative of action and inference.

Yeah. So, I'm not sure how I should understand this.

So, for instance, I want to go to the baker and buy some bread.

And then, the most informative, I mean the more unlikely thing is actually that I go to the barber and cut my hairs.

So, I would, because that has more information than going to the baker and considering what kind of bread I'm going to buy,

I go to the barber because that is more informative.

I'm not sure I understand how I should understand this.

So, these values are calculated about generative models, not about territories.

So, again, it's natural and intuitive to want to be like directly mapping from an equation or calculation to a real world territory situation.

But, keep in mind, these are ultimately numerical values calculated for a generative model.

It doesn't make sense to talk about free energy being minimized, like, for example, of a person doing translation simply or of an ant simply foraging.

It's of the generative model that these values are being calculated on.

And so, if one proposes a generative model of location behavior, so, kind of in a previous discussion, we had like a higher order, like, what location am I going to?

Barber, home, baker, you know?

And then, we have a lower level model with our footsteps.

So, that's kind of analogous to the live stream 42 SLAM robotics example.

Then, in that situation, one could ask how different policy selections were being influenced by their respective epistemic or pragmatic value.

But, it is not the case that we can directly draw from a variable to the territory.

These are like summary variables or re-weightings on our map.

And, it only makes sense to talk about these summaries on maps, not territories.

Ali?

Actually, Michael Levin has an excellent paper from 2019, The Computational Boundary of a Self, in which he has an interesting diagram showing the cognitive sphere or cognitive light cones, if you want to say.

The cognitive or intelligent agent, in which it can have an effect.

And, it's of, more or less, of its concern.

It's concerned.

So, it also integrates the, yeah, that's it.

So, for example, for humans, because of, well, or highly intelligent species or agent, because of the complexity of its goal-directed activity and agency, it results in a kind of larger cognitive boundaries.

And, for somewhat, let's say, quote unquote, simpler agents, that would result in a much smaller light cone area of cognitive boundaries.

So, yeah, depending on the complexity of the tasks that the agent is capable of doing, that territory of its cognitive effectivity would vary according to the complexity of the tasks.

So, yeah.

Yeah.

So.

So.

So.

So, these are, G is a functional that is going to sharpen our action prior according to such and such.

So, if we just wanted to pass through habit, we would just pass through habit.

And, in chapter five, in the nervous system example, this is what's reflected in the bottom here with this dopaminergic circuitry.

Where on the left side of this basal ganglia or whatever mammalian tissue it happens to be, on the left side, there's kind of like a habit pass through.

Like, E, the prior on affordance, is basically passed through.

Whereas in G, I'm sorry, on the right side, G is sharpening the policy distribution.

So, if we just want to pass through habit, we don't even need to apply G.

However, if we want to sharpen our action selection the moment to respect expected free energy in a prospective planning-like fashion, then we want to calculate G.

So, what is being calculated with G?

Again, this is an equation 2.6, but also here we're taking another look at it.

So, there's two terms.

Looking first at the pragmatic value.

And, the special cases were already mentioned and elsewhere about when one of these or the other is zero.

Okay.

So, fancy E.

It's an expectation regarding RQ, variational distribution that we control,

of O tilde observation sequence conditioned upon policy.

So, we're going to see that both terms are conditioned upon policy.

So, if there's only two affordances, then we're only going to compute two different Gs.

So, this is how we kind of prevent like spiraling and computing like all possible world futures.

We're conditioning on action.

And, we're interested in reducing our uncertainty about which actions to take.

So, that's what this first part is saying.

It's expectation of the variational distribution of upcoming observation sequences conditioned upon actions policies that we can take.

And, it is going to be a natural log to help facilitate just all kinds of things that logs facilitate.

But, it's a monotonic relationship with what is being described here.

And, this is about observations conditioned on preferences, C.

So, where observations are close to preferences, pragmatic value is high.

Where our observations are different than our preferences, pragmatic value is low.

So, rather than associating pragmatic value with a reward function, we're associating it with bounding our surprise, natural log, self-information, surprisal, of our preference slash expectation.

So, the epistemic value component, which can be kind of seen as like a sub-function with like a negative i.

But, it's interesting that also this is kind of a surprisal-like, like even pragmatic value enters into an information theoretic surprise-like formulation.

So, even our pragmatic value is still informational.

It's not, you know, tokens.

Here, we see, again, an expectation over our variational distribution.

But, here, rather than on the sequence of observations through time O tilde, it's the sequence of hidden states through time.

So, this is now looking at underlying latent states conditioned upon policy.

And, there's two, there's a difference of two entropies.

Then, here's P, which is like the kind of actual distribution, and then Q, the distribution we control.

Now, within the parentheses, O tilde in both cases, O tilde conditioned upon states minus O tilde conditioned upon policies.

So, there's definitely a few ways to unpack it, and I think we've explored some.

But, broadly, if the policy is not further reducing your uncertainty about this relationship, the epistemic value is low.

If the policy is reducing your uncertainty about S via O, then there's epistemic value.

I think there's, like, a lot of open questions that are pretty contextual.

Like, how do these implicit or tacit understandings, like, how does the rat know that going to that Q location is going to reduce its uncertainty?

How does it know the semantics of the left versus the right Q?

How does it know what, you know, this is not the whole story.

But, within the context of doing variational inference on a structurally determined model,

this is the basis of the separation between pragmatic value, defined as alignment between observations and preferences,

and epistemic value, defined as reduction of uncertainty, which is to say entropy,

of how policies sharpen our understanding of the relationship between hidden states and observations.

Michael?

Yeah.

So, both of these are expectations.

So, that is with respect to future things that might happen.

Both of these expectations are actually used in the sense of a mean.

The time horizon is implicit within pi.

Whether that is being calculated, whether G is being calculated over pi of length 1, 2, 3, etc.,

in the discrete time formalization.

But these expectations are actually, like, center of gravity,

Laplace approximation,

central limit theorem,

think expectation of a distribution,

even though it's, like, about the future.

We're actually computing, like, the center of gravity

or policies of different times.
But we're not, funny enough, like, we are,
but we aren't talking about future time.
Okay.
So, what I maybe don't really understand here
is the difference between P and Q.
I was of the understanding that P is something that we,
that is present in the presence that we experience,
and Q is something that happens in the future.
Now, I'm a little bit confused
because P and Q both relate here in this equation to the future, right?
So, how can this be understood?
The difference between P and Q is not between the present and the future.
P is, like, the underlying distribution,
and Q is our variational approximation.
So, this is, like, the actual entropy of the relationship
between observations and hidden states,
whereas this is the relationship of the entropy,
the entropy of the variational generative models relationship
between those two.
Ali?
Yeah, I think equation 2.3 might help to somehow clarify
that ambiguity between P and Q,
because in equation 2.3,
yeah, that's it,
the KL divergence between Q and P
is defined as the expectation
between the surprisal of Q and the surprisal of P.
And one interesting thing about equation 7.4
is that the epistemic value
is the expectation with respect
to the hidden states,
but the pragmatic value
is the expectation with respect to the observations.
So, we need to keep both of them in mind
because here, as Daniel just mentioned,
P is just the...
Q is the variational...

I mean, free energy,
but P is the standard probability function
we apply to a random variable.
So, that's why they have distinguished
between those two functions,
because Q has some specific definition.
So, for example, in equation 2.2,
the probability of P without any...
Yeah, that's just the simple derivation
of the probability of a random variable
being of a specific value is calculated
without taking into account Qs
or other parameters.
Nice.
Yes, the fact that pragmatic value
is associated with outcomes
is what ties closely to perceptual control theory.
And the fact that epistemic value
is associated with hidden states
is related to the
avoidable of overfitting.
We're not reducing our epistemic value
on observations.
Observations are being evaluated
for their pragmatic value.
Hidden state inferences are being evaluated
and through them,
policies are being evaluated
for their epistemic value.
Michael?
Yeah, so...
I'm a little bit confused
about this.
You said that Q
relates to the variational free energy,
but the variational free energy...
I mean, we made this distinction
between the G,

which is the expected free energy,
and the F,
which is the variational free energy,
and now inside that expected free energy
is the variational free energy.
So, yeah, just to be clear,
Q is the variational distribution.
This is the guy we control.
Q is the guy we control.
P, we don't.
Mind your P's and Q's.
Q is the one we control.
Variational free energy
is a real-time functional
that's a function of
our variational density, Q,
and incoming data.
It doesn't engage with planning.
Expected free energy
and
a host of cousins
like free energy
of the expected future
that Barron-Millage
and others have proposed.
So, this is not like the only way
to even do this energy-based method
in the future.
This
is
rather than a function
of
taking in
only Q and Y
as arguments.
Here,
the primary argument
is

the policy prior vector.
And then
the computation
of this functional
involves
Q,
the variational distribution.
But, like,
Q could be
a Gaussian distribution.
So, we'd be looking
to parameterize
the variational parameters,
you know,
the median
and the variance,
the two parameters
of a Gaussian.
and
the true
generative process
may not be
Gaussian.
But, we would be looking
just like
in a linear regression,
we'd be looking
to minimize
the sum of squares
with the L2 norm.
In this
variational inference
setting,
we'd be looking
to fit
the best Gaussian
mean and variance
given

the family
of the variational
distributions
that we had chosen.
Ali?
Yeah, so,
one other angle
can be
E is
the actual
posterior probability
that
is calculated
according
to the
actual states
we have
at our disposal.
But,
Q
is the approximate
posterior.
So, it's not
the exact
posterior
that
has been
calculated
in P.
But, yeah,
Q is
related
to variation
of free energy
in the sense
that it's deployed
in calculating
the variation

of free energy
because
for
the,
in order
for the free energy
to be
a tractable
parameter
to be
implementable
in a computational
task,
we need
to use
this approximate
variational
free energy
because otherwise
we don't
have
either
all the
information
we need
to calculate
the exact
posteriors
or
it's
too
intractable
to compute.
So,
that's the
reason
behind
using

approximate
variational
free energy.
Great.
And I think
one
kind of
Leslie Valiant,
this idea
of like
probably
approximately
correct,
which is
kind of like
when they say
almost surely
in math
but more
on the statistics
side,
if it's
probably
approximately
correct,
it's like
actually
within the
context
of
approximate
base,
that is
what
is
going to
be
not just

good enough
but like
as good
as you
can kind
of aim
for.
Okay.
Matrices
are
defined.
A
has
two
slices,
A1
and A2.
We're
going to
think of
it as
two
modalities
of A.
and
the
dimensionality
is
clear
in the
MATLAB
implementations,
it's clear
in the
PyMDP,
it's not
exactly
exactly

the same
syntax
but like
the
structure
of these
matrices
in principle
are the
same.
A1
is
describing
the
probabilistic
mapping
from
location
to
exteroceptive
cues.
So
it's
not
exactly
an
identity
matrix
but
basically
this is
the
accurate
GPS
sensor.
When
it's
at

this
location
it knows
exactly
where
it is.
So
again
you can
be like
how
does
it
get
this
implicit
knowledge
of
where
it
is?
Right.
That's
part of
the
complexity
of
cognitive
modeling
but
suffice
to say
that
we're
in
a
setting
where

the
location
is
known
without
error
but
one
can
imagine
that
there
would
be
different
outcomes
and
the
need
to
do
different
parameterizations
if
there
were
off
diagonal
elements.
A2
is
the
probability
so
they
sum
across
to

1
about
the
food
location
between
the
spatial
location
and
the
outcome
of
food.
So
semantically
this
represents
beliefs
about
food
location
in
the
A.
Here
7.5
is the
same
as
7.4
it's
just
that
here
it's
like
2

in
the
upper
left
98
on
the
off
here
we
see
98
on
the
upper
left
2
on
the
off
here
we
have
one
in
the
third
row
empty
on
the
second
here
those
two
rows
have
been

flipped
to
represent
the
black
on
the
left
side
versus
the
black
on
the
right
side
so
the
food
can be
either
on the
left
or
the
right
side
the
rat
can
either
make
the
move
to
go
to
the

left
or
the
right
side
or
can
go
to
the
queue
if it
decides
to go
for
the
queue
not the
queue
letter
but the
CUE
if it
decides
to
receive
an
informational
queue
then
it
will
find
out
where
the
food
is

again
check
out
model
stream
7.2
with
epistemic
foraging
and all
of that
sort
of
stuff
okay
these
slices
of B
are
describing
movement
affordances
because
again
with respect
to the
hidden
state
being
location
B
describes
how
different
policy
selection
bears
upon

hidden
states
changing
through
time
C
is
describing
the
relative
preference
for
different
outcomes
so
the
C1
vector
is
about
the
first
modality
five
of them
that's
locations
and there's
a negative
for the
starting
location
so that's
like a
little bit
of like a
kind of
get going

because if it
preferred to
start location
it might just
stay there
then in the
second
modality
that's
referring to
this
shorter
vector
of these
one
two
three
and that's
these
one
two
three
in
various
modeling
settings
we've
already
started to
see that
like
the C
scaling
matters
a lot
and so
that's one
reason why

I think
it's
tentative
or hypothetical
to say
that simply
like equation
7.4
resolves
explore
exploit
because
it sets
up
the space
for explore
exploit
to be
framed
absolutely
it does
however
depending on
how the
variables are
parameterized
you can end
up with an
agent who
still goes
for the
pragmatic
value
every time
or who
still goes
for the
information

value
every time
so
in terms
of
outcomes
it's not
simply
resolved
by knowing
what equation
2.5
is
but
rather
this
opens up
a space
of modeling
in which
adaptive
solutions
can be
reached
so if
this was
6,000
then
0
6,000
0
it's like
reward is
so good
it's going
to be
pursued
if this

was flat
0
0
0
then it
would be
pure
epistemic
information
so
that's
why
it's
so
important
that
we
develop
statistical
power
methods
and
parameter
sweeps
across
simulations
because
just
making
one
T-maze
and saying
oh look
the rat
does this
or that
it's
kind

of
like
okay
D
also
is
described
for the
two
modalities
and
it
describes
the
priors
on
the
two
there's
a
belief
about
the
first
location
I'm
actually
just
just
I'm
a little
bit
curious
why
there's
five
elements
here

but
only
four
elements
here
and
then
in the
second
modality
there
which
again
is
referring
maybe
this
is
like
a
minus
there's
one
fewer
element
here
because
there's
only
two
here
here's
like
reflecting
an
equal
prior
over

where
the
food
is
so
a
priori
is
50-50
left or
right
so
do
you
do
a
50-50
bet
left or
right
on the
food
or
do
you
go
to
get
the
epistemic
queue
which
of course
reduces
your
uncertainty
about
where

the
food
is
now
we're
going
to
dive
a
little
bit
deeper
into
epistemic
value
itself
so
in
7.4
they're
saying
we're
going to
go ahead
and call
just
the
epistemic
value
decomposition
term
i
equation
7.8
so
here
is
just

a
separation
of
as
it
was
written
in
the
prior
equation
that
is
going
to
be
rewritten
in
terms
of
a
KL
divergence
q
s
and
pi
q
and
s
and
pi
they're
the
same
on
both
sides

here's
pos
here's
q
o
pi
and
it's
it's
definitely
worthwhile
i think
for us
to write
out
what
these
verbally
mean
because
they
have
very
i think
interesting
implications
for
again
like
what
is
the
epistemic
value
and
then
here
is

with
this
semicolon
being
noted
this
is
reflecting
a
pure
information
gain
salience
Bayesian
surprise
imperative
here's
some
simulations
showing
at like
time
two
it goes
to the
get the
information
and then
it
proceeds
to the
left
Ali
yeah
just
about
your
previous

remark
about
why
c
has
five
elements
and
d
has
four
elements
because
c
actually
corresponds
with
the
elements
of
a
matrices
so
because
a
matrices
actually
describe
the
hidden
the
states
that
the
environment
is
dealing
with

but
because
d
corresponds
with b
matrix
and
because b
matrix
is a
transition
matrix
that's
why
we don't
need
to have
a
redundant
additional
element
for that
so
d
being
priors
yet
maps
onto
b
matrix
thanks

and
right
now
like
you

have
to
specify
the
tensor
dimensionality
and
add
some
labels
but
they're
developing
a lot
of
functions
in
pi
mdp
and
people
who
work
with
it
can
definitely
have
a lot
of
feedback
on
slightly
more
intuitive
and
scalable
ways

to
specify
dimensionality
which
is
currently
one of
the
rate
limiting
steps
in
pi
mdp
application
okay
just
getting
through
seven
and
then
just
so
we
can
look
ahead
to
eight
continuous
time
so
precision
negative
entropy
neg
entropy

this
is
the
high
temperature
and
low
temperature
high
temperature
differences
are
erased
low
precision
low
temperature
absolute
zero
high
precision
differences
are
accentuated
that
has
technical
usage
especially
in
certain
statistical
distributions
isacade
paradigm
par
and
frist

in
2017
eye
circadian
is
while
our
eyes
may
dwell
on
what
we
may
associate
with
pragmatically
value
a bull
broadly
speaking
eye
circadian
is driven
by
information
gain
so
it's a
great
place
to study
how
pragmatic
semantics
at a
higher
level

are in
a
predictive
processing
type
relationship
with
eye
circadian
at a
lower
mechanical
level
with
this
level
almost
surely
being driven
by information
gain
learning
and novelty
here
one
kind of
question
that was
arising
was like
why is
there a
dashed
line
between
only
here
but

apart
from
that
priors
are being
provided
on
a and
b
so
that's
sort of
like
again
and it's
like a
little bit
like
why is
b
over
here
so
it's
about
the
topology
not
the
geometry
in this
situation
what's
happening
is that
a prior
is being
provided

on
a
so
that
these
a and
b
in
principle
can be
learnable
how that
plays out
and how
that
influences
the
pseudocode
of the
action
perception
loop
pi
mdp
is very
clear
about
that
conjugate
priors
this is
facilitating
like
amongst
other
abilities
the
the

dirichlet
conjugate
prior
for the
categorical
distribution
so if we
have probability
distribution
over categorical
outcomes
is it
hot or
cold
then
the
dirichlet
can be
used
in an
urn
counting
way
like
every time
you observe
a warm
day or
a cold
day
you just
put in
a ball
of that
color
into the
urn
and then

it could
be
two
and
two
and
then
the
next
ball
is
very
influential
it
could
be
a
thousand
and
a
thousand
and
the
next
ball
is
not
so
influential
and
that
turns
out to
be
what's
called
a
conjugate

prior
that
enables
the
categorical
distribution
to
stay
as a
classical
probability
distribution
Kolomogorov's
axioms
and all
of that
while also
allowing
the
conjugate
prior
to
have
this
like
counting

learning
by
counting
strategy
inferential
approach
to
learning
this
is a
difference

with
many
other
areas
they're
proposing
learning
schemes
that
may or
may
not
be
biologically
relevant
whereas
a lot
of
work
has
gone
into
understanding
how
the
perceptual
or
fast
inference
and
learning
or
slower
parameter
learning
processes
are
unified

but
of
distinct
time
scales
and
active
inference
here
in
7.11
equation
they
are
going
to
delve
more
into
learning
as
active
yes
it's
about
policy
everything
is being
conditioned
upon
policy
there's
a lot
that one
can say
about
the
integration

of
action
so
this
maze
learning
task
is
fundamentally
intertwined
with
the
action
selection
the
agents
updating
of
their
likelihood
is
related
to
their
action
selection
amidst
uncertainty
they're
not
seen
as
two
separated
phases
here's
where
they

just
drop
some
threads
and
further
literature
maybe
further
versions
of the
textbook
are
going
to
come
into
more
detail
hierarchical
deep
inference
nested
modeling
something
that people
talk
and think
about
a lot
but
here
we
see
the
figure
4.3
or

figure
7.3
the
rake
and
the
policy
selection
fractal
it's
a
rake
of
rakes
and
structure
learning
and
Bayesian
model
reduction
just
some
model
comparison
and
analysis
techniques
that
support
Bayesian
statistics
another
example
involving
a
hierarchical
inference

model
is
described
and
that
has
been
used
in
the
context
of
reading
and
other
visual
inference
tasks
that's
chapter
7
let's
quickly
look
to
chapter
8
chapter
7
was
discrete
time
generative
models
chapter
8
is
going

to
be
continuous
time
generative
models
models
the
main
setting
in
which
discrete
time
models
have
been
studied
is
central
or
cognitive
decision
making
categorical
decision
making
in
contrast
the
major
setting
in
which
continuous
time
active
inference

models
to
this
time
have
been
used
in
is
in
continuous
movement
or
motor
control
tasks
so
like
in
live
stream
46
active
models
do
not
contradict
folk
psychology
with
Alex
Kiefer
Ryan
Smith
and
Maxwell
Ramstad
they

talk
about
motor
AI
MAI
and
decision
active
inference
DAI
here
there's
going
to be
a
proto
Bayesian
physics
like
representation
that's
going to
enable
us
to do
continuous
things
specifically
the
separation
of
a
data
observation
sequence
in
terms
of

an
underlying
function
and a
fluctuation
stochastic
term
and
then
a
latent
state
derivative
with a
little
dot
change
in latent
states
as a
function
of the
current
status
of the
hidden
state
and
slowly
varying
causes
which
are
going
to
be
playing
the

role
of
policies
in
continuous
time
formalization
and
again
partitioning
out
of
stochastic
fluctuations
so
actually
this part
is
guaranteed
to have
zero
mean
and that's
what
facilitates
Bayesian
mechanics
a lot
of other
statistics
again
just like
quickly
looking
through
it
this
y

g
x
and
f
relationships
continue
to be
built
up
to
again
talk
and
this
is
very
SPM
like
here's
our
actual
sensor
readers
readings
from
the
eeg
fmri
meg
their
function
of
neural
activity
neural
activity
is a
hidden

latent
state
it is
a
function
of
current
neural
activity
and
attention
for
example
that
physics
analogy
in the
continuous
state
space
can be
extended
into
Newtonian
dynamics
and as
Dalton
and
Bayesian
mechanics
have
laid out
in the
last year
or two
it goes
way way
further

than
this
but
this
is
a
spring
so
this
is
Hooke's
law
as
a
mere
active
inference
entity
this
is
like
a
boring
active
entity
that
just
dampens
but
it's
just
to
show
that
we
can
understand
the

generalized
coordinates
of
motion
which
is
to
say
position
velocity
acceleration
snap
crackle
pop
etc
etc
etc
generalized
within
this
continuous
time
framework
dynamical
systems
often
which are
conditioned
to be
smooth
and
differentiable
are
amenable
to
continuous
time
formalizations

in
discrete
time
we
say
well
we
have
1
2
and
3
and
then
we
are
going
to
be
moving
to
one
of
the
locations
in
the
grid
world
and
we
have
a
transition
matrix
over
grid
world

in
the
continuous
time
setting
here
this
is
the
spinal
cord
with
the
butterfly
here's
the
sensory
data
coming
in
y
and
then
a
descending
hidden
state
inference
and
error
deviation
that's
being
resolved
through
policy
selection
in the

continuous
space
so
this
is
hearkening
back
to
chapter
five
and
the
spinal
cord
reflex
how
continuous
time
generative
models
can
be
applied
to
continuous
spinal
motor
reflexes
as
opposed
to
the
usually
more
centrally
associated
discrete
categorical

decision
making
DA
AI
there's
a
sidebar
on
precision
attention
and
attenuation
there's
the
discussion
of
latke
volterra
which
is
also
called
predator
prey
or
winnerless
dynamics
like
you
can't
win
in
this
ecology
there's
also
there
are

ecologies
that
collapse
of
course
but
activity
dynamics
amongst
brain
regions
have
been
modeled
for
decades
as
these
winnerless
competitions
also
called
neural
darwinism
and that's
relating to
fristin's
lineage
with
edelman
more
examples
of how
these
kinds of
winnerless
competitions
can be

used in
the context
of
continuous
time
generative
models
to generate
smooth
action
sequences
like this
sort of
gibberish

cursive
writing
learning
is
approached
in a
continuous
model
everything
in chapter
eight
is kind
of being
contrasted
or juxtaposed
with what
we saw
in chapter
seven
in the
discrete
time
they

introduced
the
Lorenz
attractor
which is
unpacked
in
significantly
more
detail
in
live
stream
32
on
Markov
blankets
and
stochastic
chaos
it is
basically
this
example
with
the
Lorenz
attractor
which is
showing
that
generalized
synchrony
which isn't
lockstep
but rather
mutual
information

amongst
couple
dynamical
systems
can happen
even when
those couple
dynamical
systems are
Lorenz
attractors
they don't
have to
be like
both
like simple
linear
systems
so that's
like good
to know
and they've
used that
to simulate
birdsong
where before
learning
they're kind
of like
on and off
a synchronization
manifold
and then after
learning
they've
sharpened
their
joint

distribution
again
this begs
further
questions
how do
they know
the semantics
of the song
and how
did they
learn the
song
but this
is just
an example
at this
level of
complexity
8.5
hybrid
models
here
we have
the rake
within a
rake
but notice
that the
outer rake
there should
be a line
connecting
this g
to pi
the outer
rake
is

d
a
i
here we
see
hidden
states
through
time
t
minus
one
now
future
tale of
two
densities
observations
through
time
classic
d
a
i
chapter
seven
up top
but the
nested
model
has the
same
structure
figure
4.3
but here
we see
the

position
velocity
acceleration
generalized
coordinates
pattern
occurring
so this
is called
a hybrid
model
because the
discrete
time
model
is
nesting
a
continuous
time
model
so this
has been
used to
model
things like
where do
I want
to look
from a
categorical
perspective
and then
here's the
oculomotor
saccades
at a
continuous

model
that's
nested
within
that
more
equations
i
saccade
model
more
equations
summary
and
key
advances
in
continuous
time
models
end
of the
chapter
all
right
thank
you
we'll
take a
break
and
then
begin
with
the
next
cohort
in

just

two

minutes